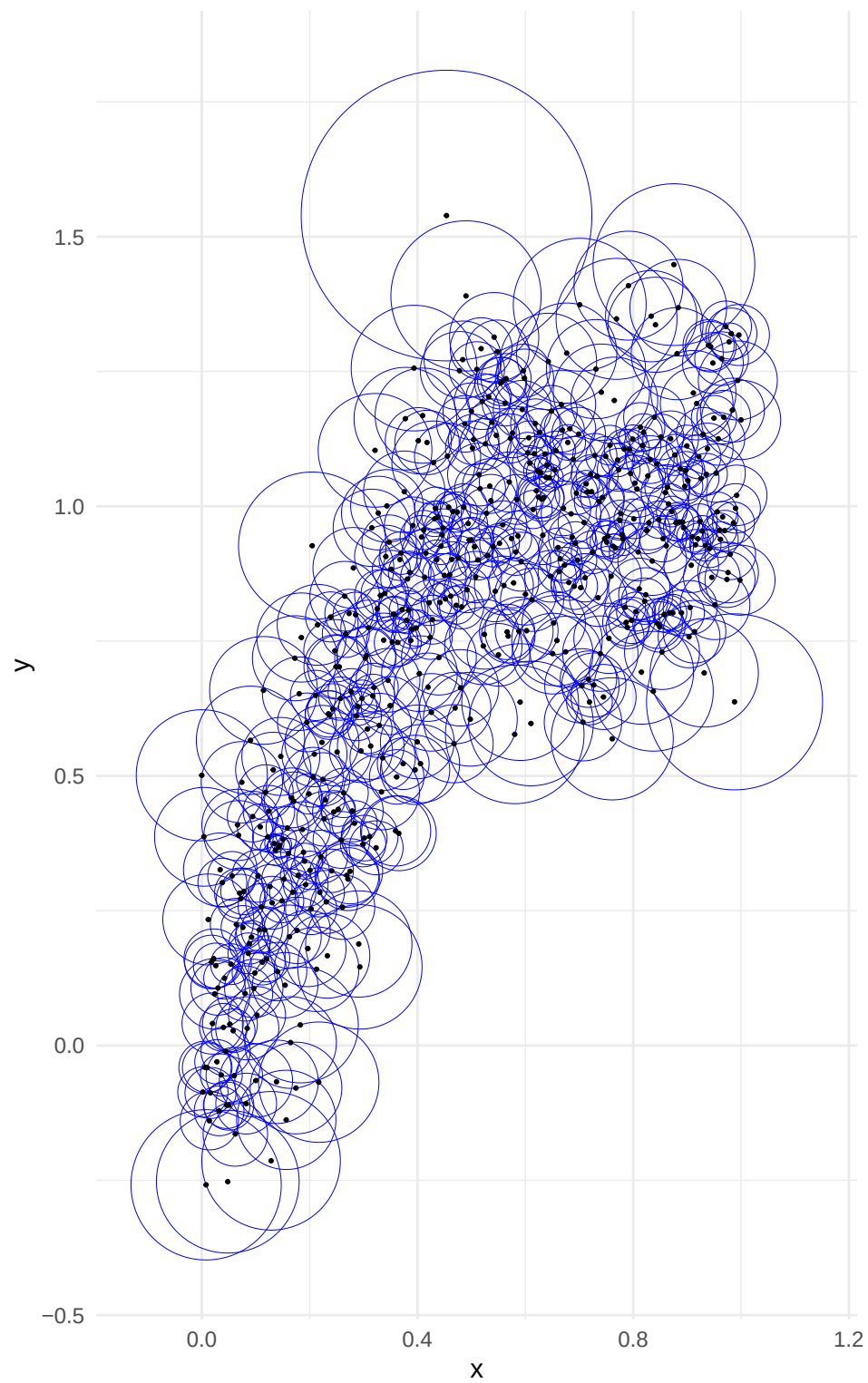
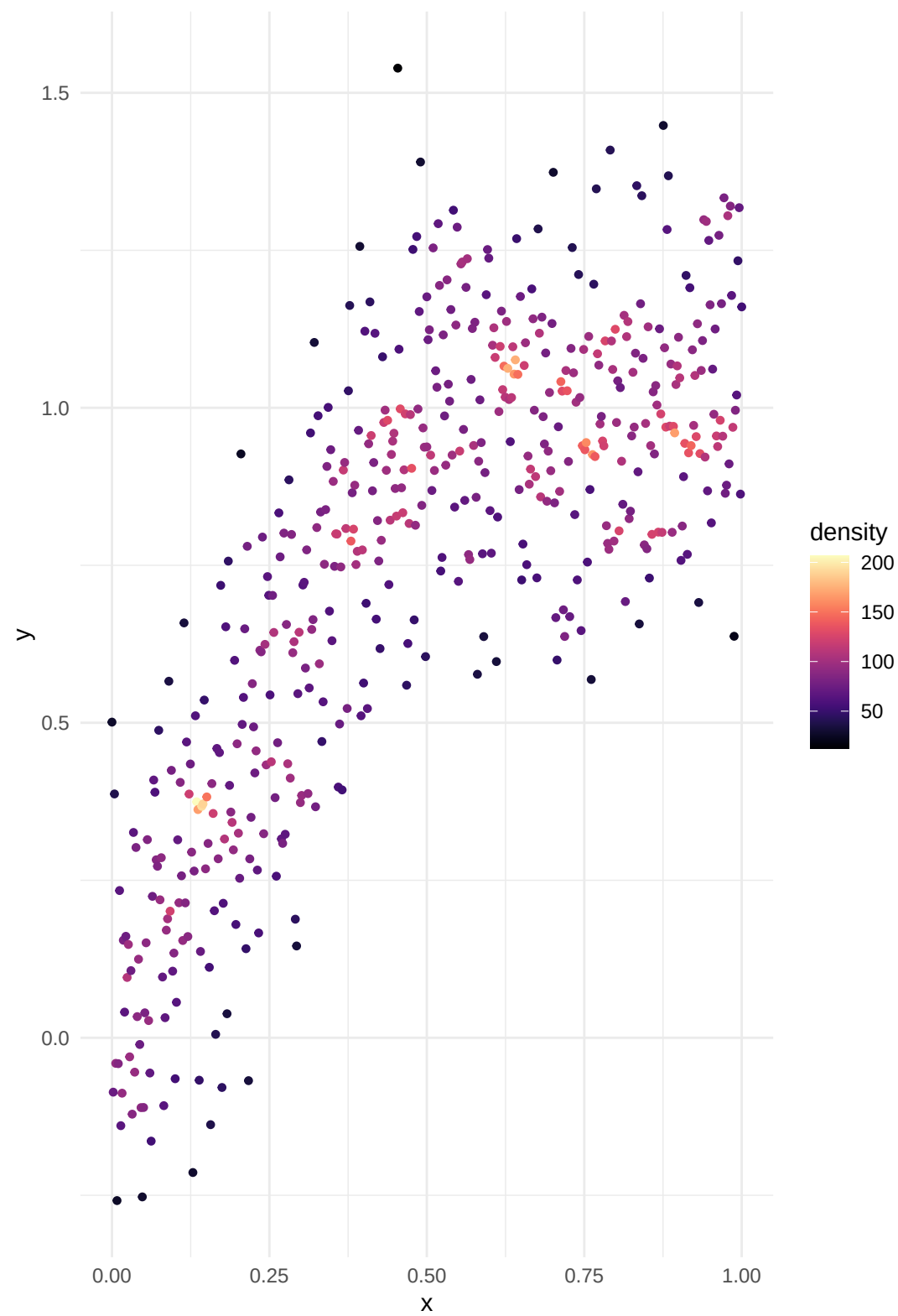


All Adaptive Spheres

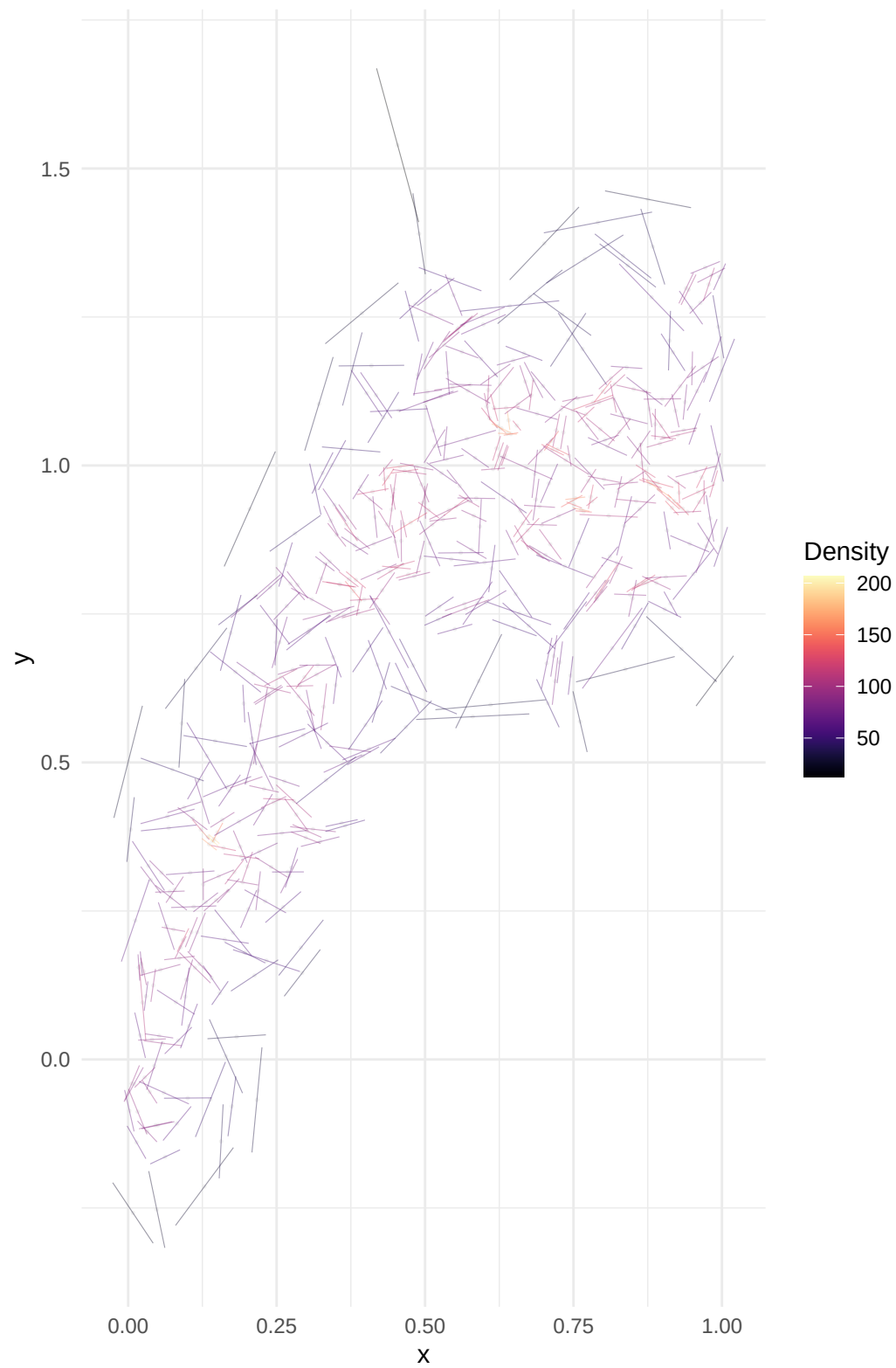
Visualizing local neighborhood size (radius) for every point.



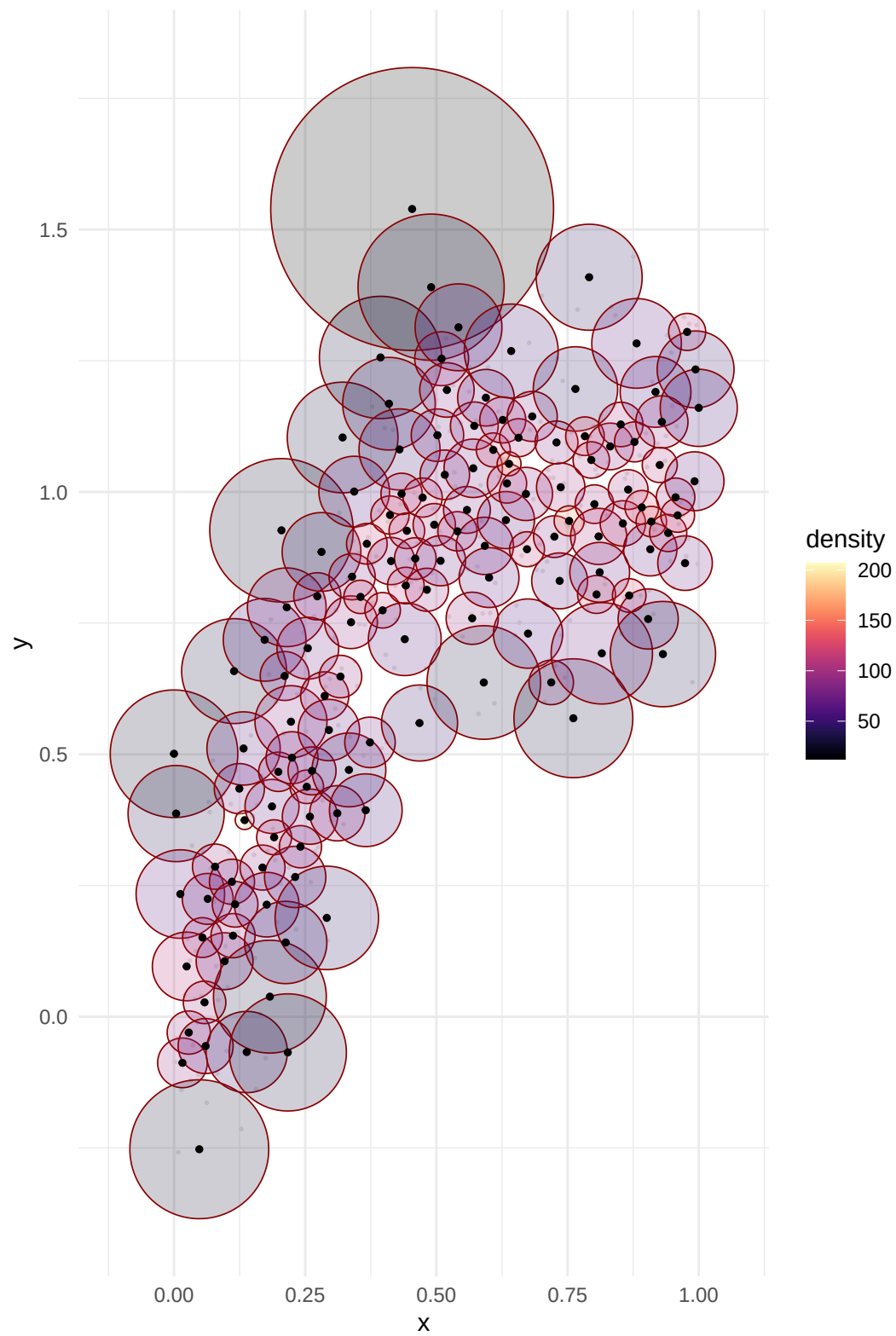
Point Density Distribution



Global Tangent Field (PC1)

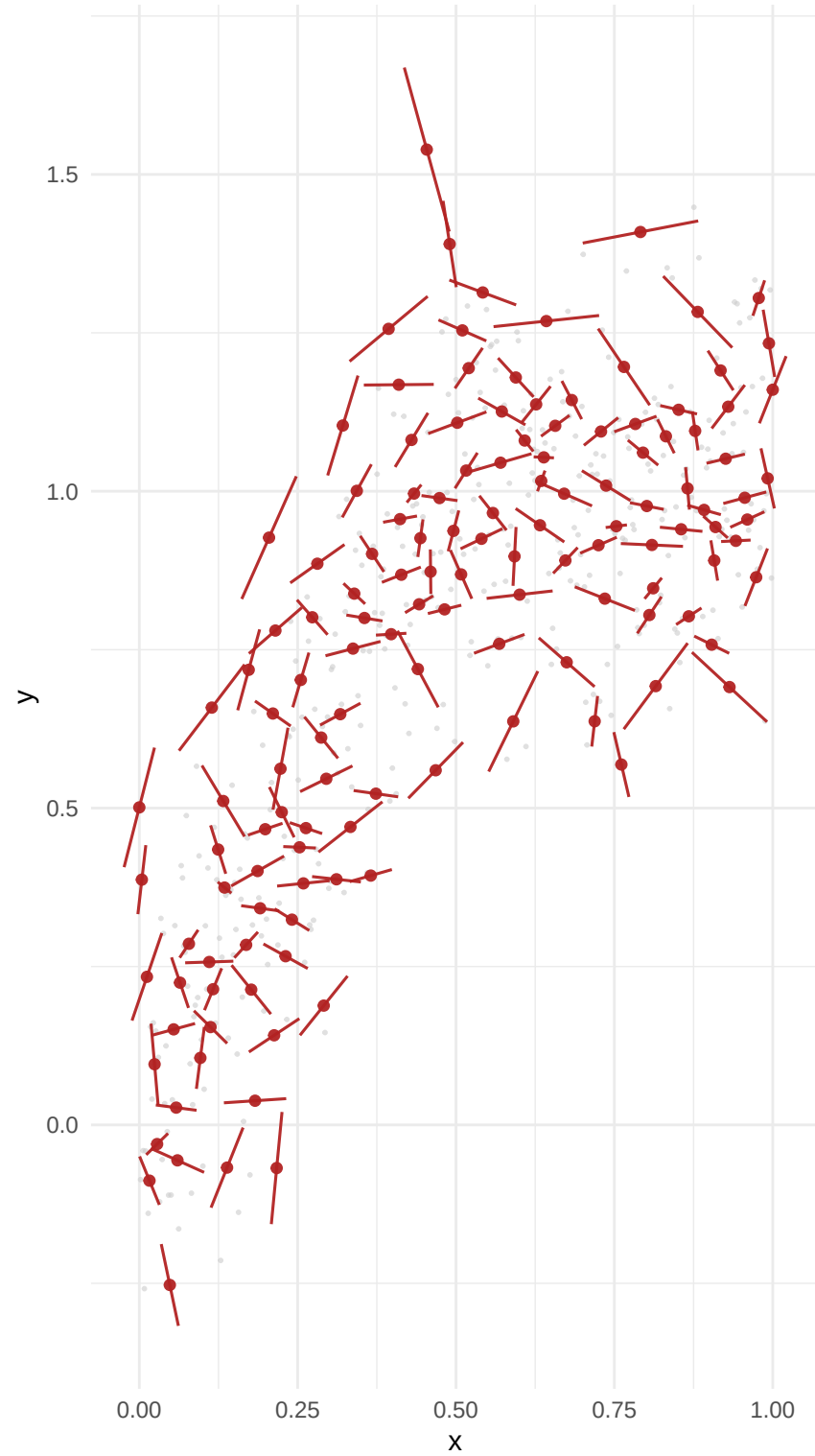


Atlas Selection (Anchors)



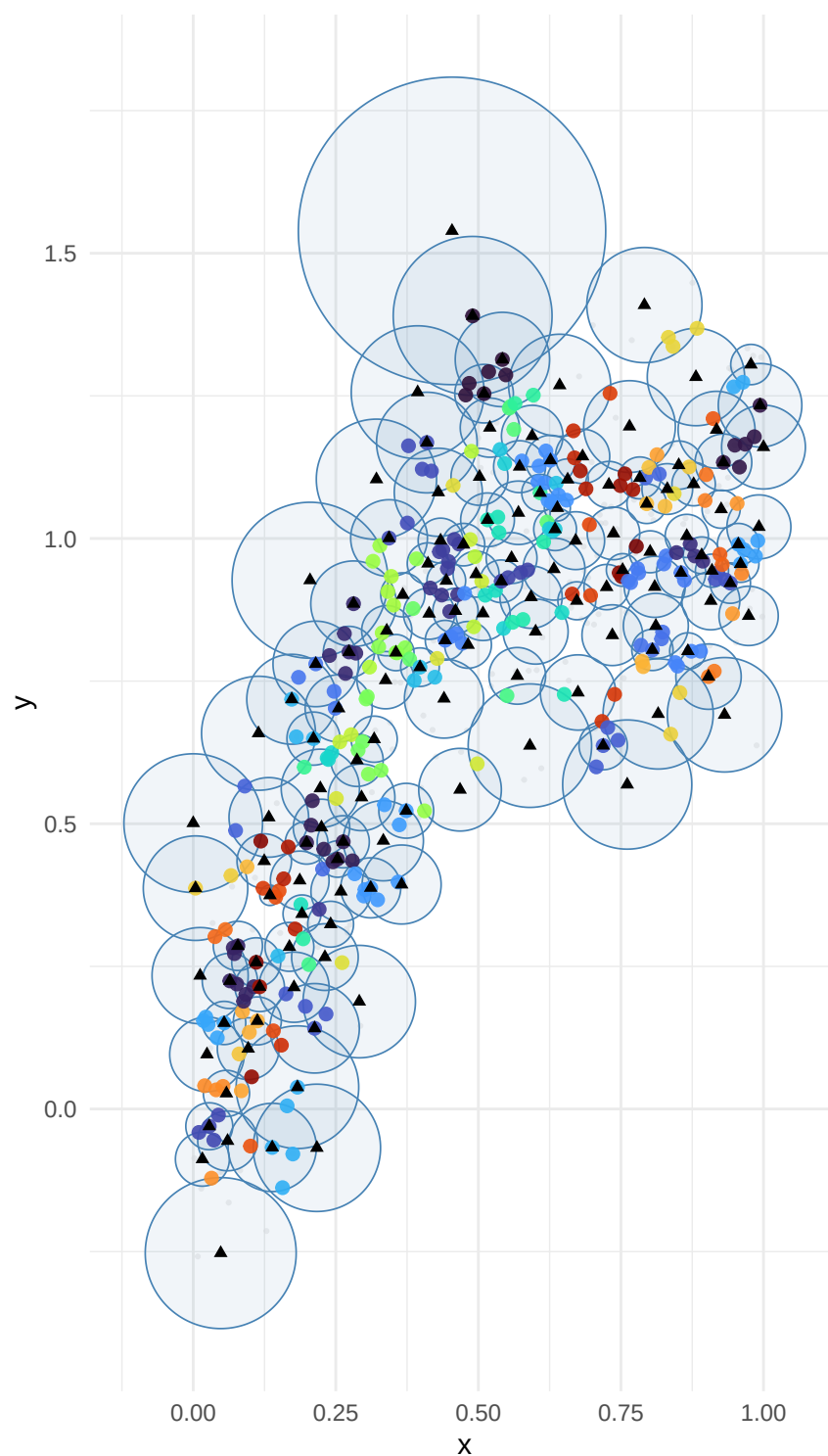
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.

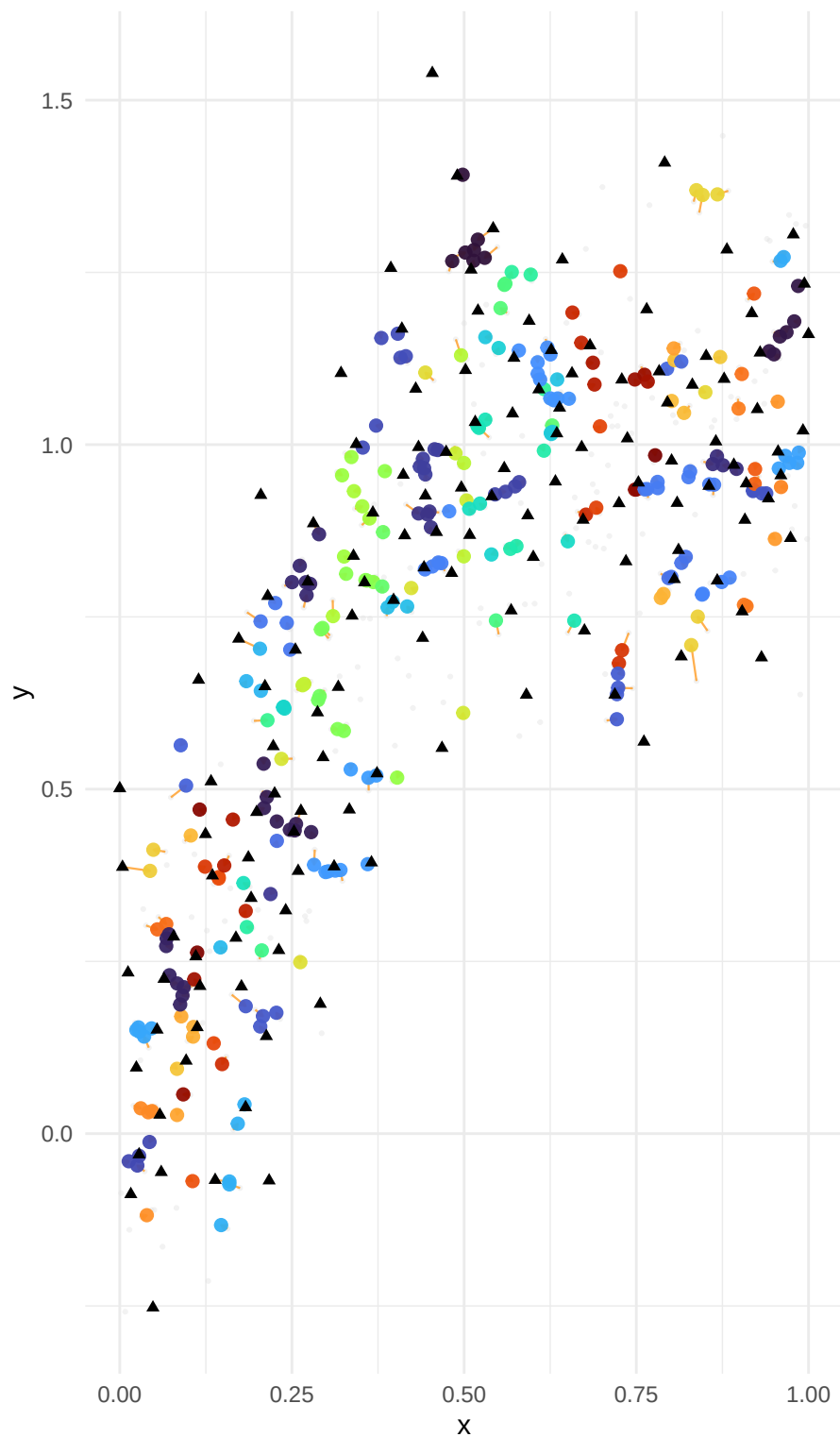


Region

1	21	41	61	81	101	121
2	22	42	62	82	102	122
3	23	43	63	83	103	123
4	24	44	64	84	104	124
5	25	45	65	85	105	125
6	26	46	66	86	106	126
7	27	47	67	87	107	127
8	28	48	68	88	108	128
9	29	49	69	89	109	129
10	30	50	70	90	110	130
11	31	51	71	91	111	131
12	32	52	72	92	112	132
13	33	53	73	93	113	133
14	34	54	74	94	114	134
15	35	55	75	95	115	
16	36	56	76	96	116	
17	37	57	77	97	117	
18	38	58	78	98	118	
19	39	59	79	99	119	
20	40	60	80	100	120	

Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

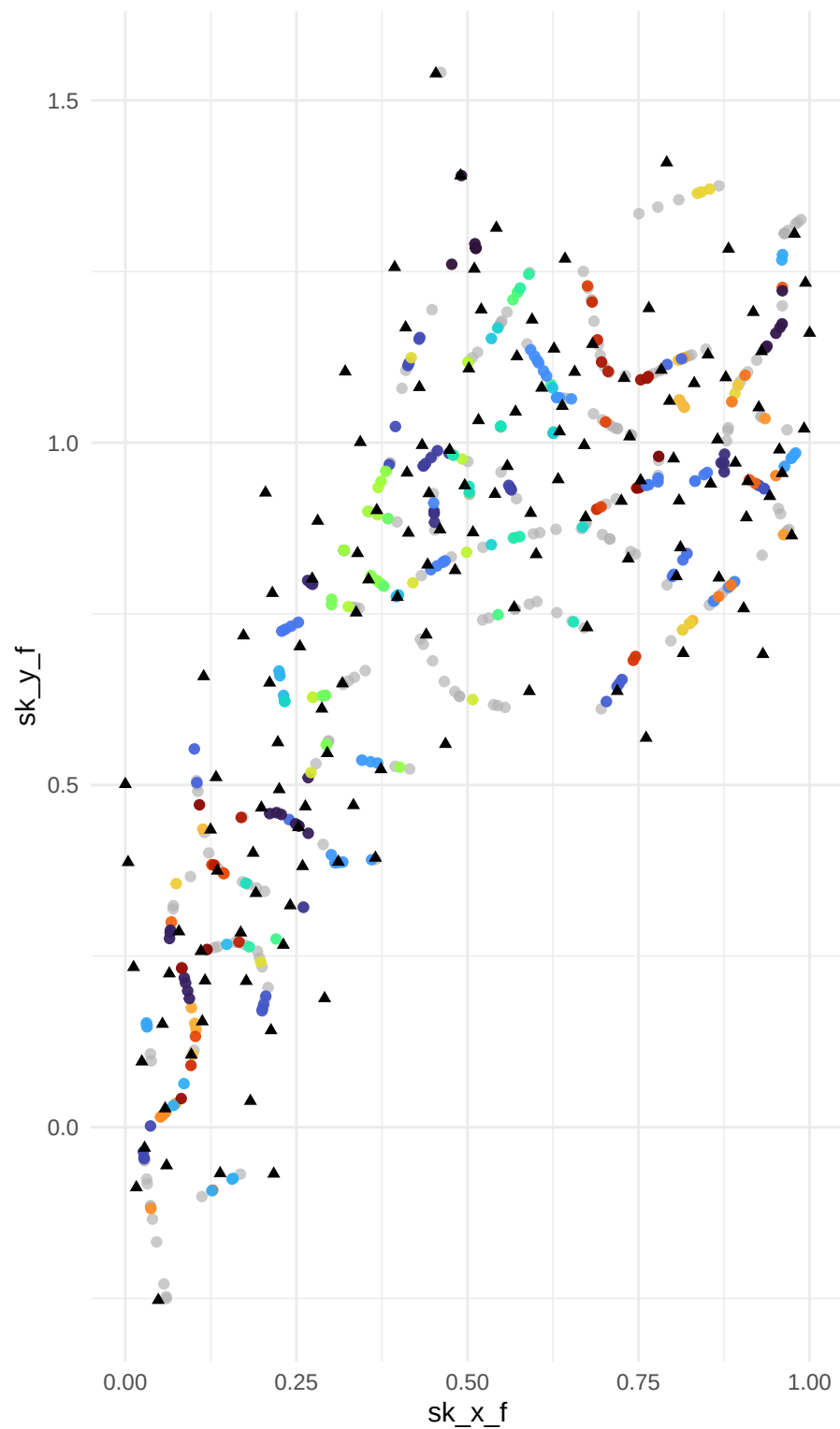


Region

1	21	41	61	81	101	121
2	22	42	62	82	102	122
3	23	43	63	83	103	123
4	24	44	64	84	104	124
5	25	45	65	85	105	125
6	26	46	66	86	106	126
7	27	47	67	87	107	127
8	28	48	68	88	108	128
9	29	49	69	89	109	129
10	30	50	70	90	110	130
11	31	51	71	91	111	131
12	32	52	72	92	112	132
13	33	53	73	93	113	133
14	34	54	74	94	114	134
15	35	55	75	95	115	
16	36	56	76	96	116	
17	37	57	77	97	117	
18	38	58	78	98	118	
19	39	59	79	99	119	
20	40	60	80	100	120	

Final Stitched Backbone (Converged)

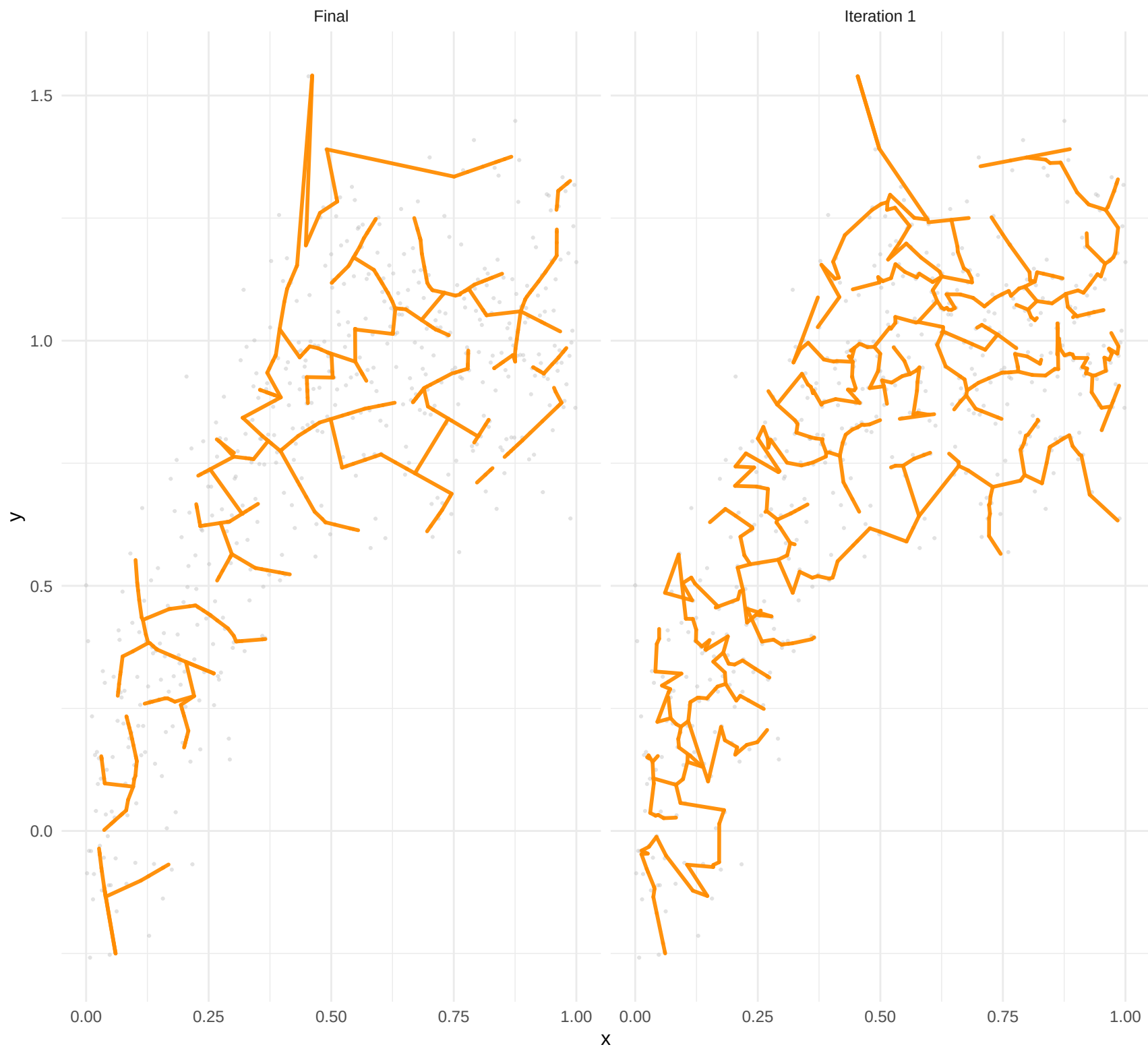
Final stitched points



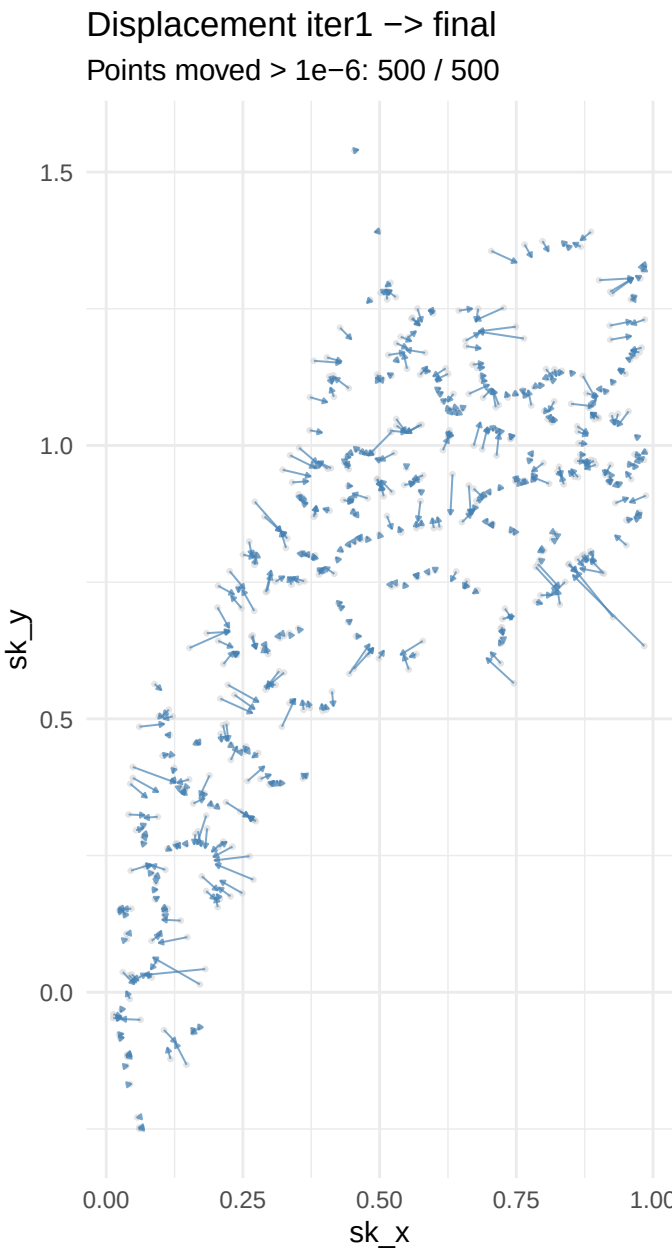
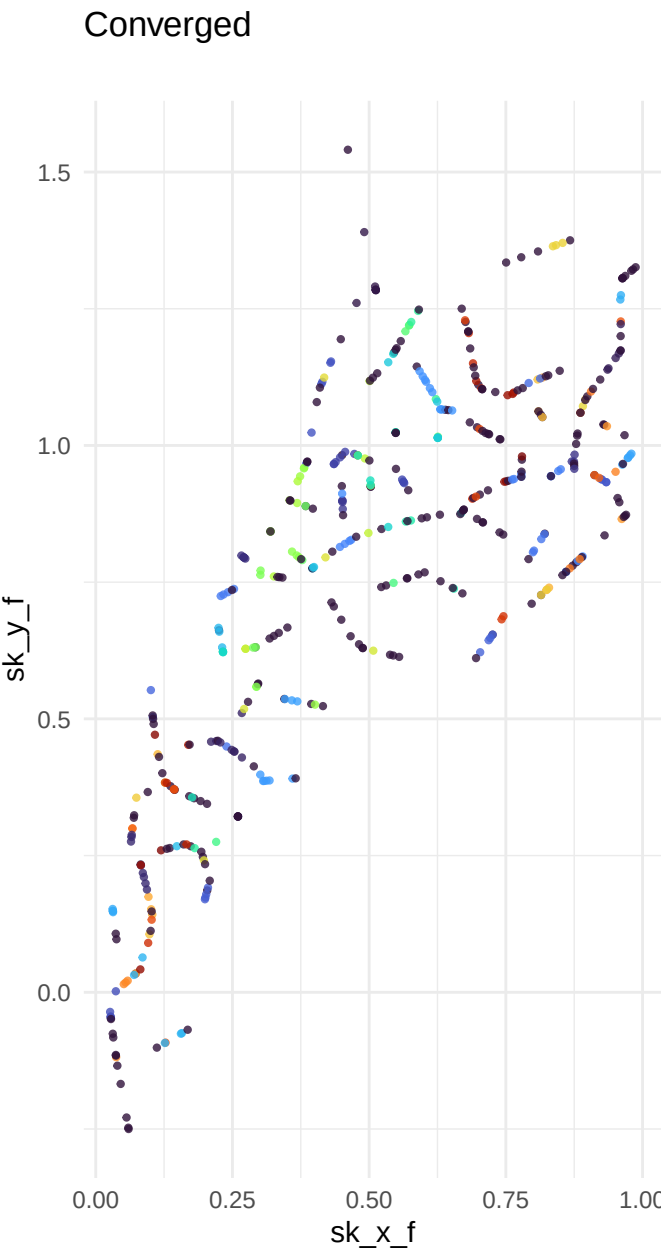
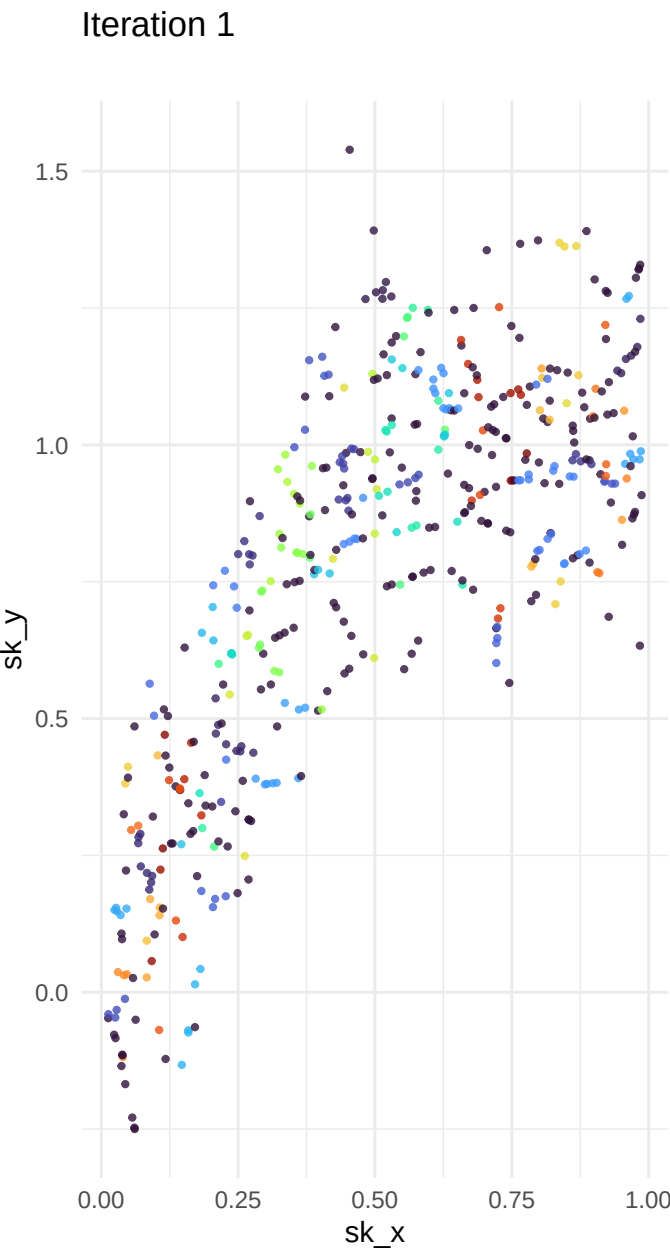
Region

1	21	41	61	81	101	121
2	22	42	62	82	102	122
3	23	43	63	83	103	123
4	24	44	64	84	104	124
5	25	45	65	85	105	125
6	26	46	66	86	106	126
7	27	47	67	87	107	127
8	28	48	68	88	108	128
9	29	49	69	89	109	129
10	30	50	70	90	110	130
11	31	51	71	91	111	131
12	32	52	72	92	112	132
13	33	53	73	93	113	133
14	34	54	74	94	114	134
15	35	55	75	95	115	
16	36	56	76	96	116	
17	37	57	77	97	117	
18	38	58	78	98	118	
19	39	59	79	99	119	
20	40	60	80	100	120	

Backbone Edges



Convergence: iteration 1 vs final



Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01